

KITABU QUEST

P5

MATHEMATICS



Cover scene

learners using counters and measuring tools with a friendly mountain gorilla mascot

● Clear lessons

● Worked examples

● Fun activities

● Real-life problems

GRADE

5

Title Page

KITABU QUEST P5 Mathematics

Rwanda REB-CBC learner book

Mascot: mountain gorilla

This KITABU QUEST book turns the curriculum into short lessons, worked examples, guided activities, practice, review, and home links for Rwanda learners.

How To Use This Book

This book helps P5 learners study Mathematics one teachable topic at a time.

1. Start with the inquiry question and say what you already know.
2. Read the Learn section slowly and mark new words.
3. Try the worked example before reading the full solution.
4. Complete the activity and practice tasks.
5. Correct your answer using the success criteria.
6. Ask for support when your answer is still unclear.

Subject method: Use objects, drawings, number sentences, and worked examples before independent practice.

Learning Skills In This Book

Each topic builds knowledge, skill, values, communication, collaboration, critical thinking, creativity, digital awareness, and self-confidence.

Study actively: speak answers aloud, draw when useful, show your working, cite evidence, use local examples, and correct mistakes after feedback.

A strong answer is specific. It names the idea, gives evidence or method, applies it to the task, and checks whether it answers the question.

Table Of Contents

Use this table to move through the book one topic at a time. Each topic has a lesson opener, clear teaching, a model or example, guided work, practice, and checks for understanding.

1. Reading, writing, comparing and calculating whole numbers up to 1,000,000
2. Addition and subtraction of integers
3. Prime factorization and divisibility tests
4. Equivalent fractions and operations
5. Multiplication and division of decimal numbers
6. Application of direct proportions
7. Solving problems involving measurements of length, capacity and mass
8. Solving problems involving time intervals
9. Money and its financial applications
10. Sequences that include whole numbers, fractions and decimals
11. Constructing angles and describing angles formed by intersecting lines
12. Interpreting and constructing scale drawings
13. Calculating circumference of a circle, surface area and volume of cubes and cuboids
14. Data handling

As you finish each topic, return to the success criteria and correct one answer before moving on.

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Lesson Opener

Learning focus: Reading, writing, comparing and calculating whole numbers up to 1,000,000

Country link: a Rwandan school, market, farming, building, transport, or data problem for Reading, writing, comparing and calculating whole numbers up to 1,000,000.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- read, write, compare and make calculations on whole numbers up to 1 000 000
- identify and name place values up to 1,000,000 in both spoken and written forms
- correctly read and say written numbers in English
- explain the concept and process of addition of 3 or more whole numbers of 6 digits with or without carrying
- compare numbers of any size using place values with a general
- write the given number accurately in word

Inquiry questions:

- How can reading, writing, comparing and calculating whole numbers up to 1,000,000 help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Learn

Reading, writing, comparing and calculating whole numbers up to 1,000,000 teaches you to read, write, compare, and calculate with large whole numbers.

Core idea:

- A digit has value because of its position: ones, tens, hundreds, thousands, ten-thousands, and hundred-thousands.
- Compare numbers from left to right, starting with the greatest place value.
- For addition and subtraction, align place values before calculating.
- For multiplication or division, check whether the answer should be bigger or smaller than the starting number.

Rwanda example: class enrolment, harvest records, market stock, and school budget totals can all use whole-number operations.

Key words:

- Reading, writing, comparing and calculating whole numbers up to 1,000,000
- Reading
- writing
- comparing
- calculating
- whole
- numbers
- 1,000,000

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Worked Example

Worked example for Reading, writing, comparing and calculating whole numbers up to 1,000,000: A school library has 23,475 storybooks and 18,320 revision books. How many books are there altogether?

1. Align place values.
2. Add ones, tens, hundreds, thousands, and ten-thousands.
3. $23,475 + 18,320 = 41,795$.
4. Estimate: $23,000 + 18,000 = 41,000$, so 41,795 is reasonable.

Answer: 41,795 books.

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Vocabulary And Study Skill

Use these words and study moves before you practise Reading, writing, comparing and calculating whole numbers up to 1,000,000. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Reading, writing, comparing and calculating whole numbers up to 1,000,000
- Reading
- writing
- comparing
- calculating
- whole
- numbers
- 1,000,000

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Reading, writing, comparing and calculating whole numbers up to 1,000,000 without a clear example, method, or evidence.

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Guided Activity

Guided activity for Reading, writing, comparing and calculating whole numbers up to 1,000,000:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Practice And Correction

Practice for Reading, writing, comparing and calculating whole numbers up to 1,000,000:

Main outcome: read, write, compare and make calculations on whole numbers up to 1 000 000.

1. Solve one reading, writing, comparing and calculating whole numbers up to 1,000,000 question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to reading, writing, comparing and calculating whole numbers up to 1,000,000 and solve it.

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Reading, writing, comparing and calculating whole numbers up to 1,000,000.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Outcome Check

Outcome check for Reading, writing, comparing and calculating whole numbers up to 1,000,000:
Read, write, compare and make calculations on whole numbers up to 1 000 000.

1. Add 34,785 and 21,406.
2. Estimate the answer to check it.
3. Explain one place-value step.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Outcome Check

Outcome check for Reading, writing, comparing and calculating whole numbers up to 1,000,000: Identify and name place values up to 1,000,000 in both spoken and written forms.

1. Add 34,785 and 21,406.
2. Estimate the answer to check it.
3. Explain one place-value step.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Addition and subtraction of integers: Lesson Opener

Learning focus: Addition and subtraction of integers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Addition and subtraction of integers.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems related to addition and subtraction of negative and positive integers
- explain that when two numbers have opposite signs, they are located on opposite sides of zero on the number line
- describe the concept of addition and subtraction of integers
- add and subtract integers using a number line, counters, or without them
- calculate the additive inverse of any integer
- solve mathematical problems involving addition and subtraction of integers

Inquiry questions:

- How can addition and subtraction of integers help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Addition and subtraction of integers: Learn

Addition and subtraction of integers teaches numbers below, at, and above zero.

Core idea:

- Positive numbers are greater than zero.
- Negative numbers are less than zero.
- Zero is the turning point.
- A number line helps you compare and order integers.

Rwanda example: temperature changes, money owed, points gained or lost in a game, and height above or below a reference point can use integers.

Key words:

- Addition and subtraction of integers
- Addition
- subtraction
- integers
- method
- model
- unit
- answer

Addition and subtraction of integers: Worked Example

Worked example for Addition and subtraction of integers: The morning temperature is 3 degrees above zero. At night it drops by 7 degrees. What is the night temperature?

1. Start at +3 on a number line.
2. Move 7 steps left because the temperature drops.
3. $+3 - 7 = -4$.

Answer: 4 degrees below zero.

Addition and subtraction of integers: Vocabulary And Study Skill

Use these words and study moves before you practise Addition and subtraction of integers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Addition and subtraction of integers
- Addition
- subtraction
- integers
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Addition and subtraction of integers without a clear example, method, or evidence.

Addition and subtraction of integers: Guided Activity

Guided activity for Addition and subtraction of integers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Addition and subtraction of integers: Practice And Correction

Practice for Addition and subtraction of integers:

Main outcome: solve problems related to addition and subtraction of negative and positive integers.

1. Solve one addition and subtraction of integers question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to addition and subtraction of integers and solve it.

Addition and subtraction of integers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Addition and subtraction of integers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Addition and subtraction of integers: Outcome Check

Outcome check for Addition and subtraction of integers: Solve problems related to addition and subtraction of negative and positive integers.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Addition and subtraction of integers: Outcome Check

Outcome check for Addition and subtraction of integers: Explain that when two numbers have opposite signs, they are located on opposite sides of zero on the number line.

1. Place -3, 0, and +4 on a number line.
2. Which number is greatest?
3. Find the distance from -3 to +4.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Prime factorization and divisibility tests: Lesson Opener

Learning focus: Prime factorization and divisibility tests

Country link: a Rwandan school, market, farming, building, transport, or data problem for Prime factorization and divisibility tests.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- prime factorize, exploit the rule of divisibility tests less than 10, find the Lowest Common Multiple (LCM) and the Greatest Common Factor (GCF) of whole numbers
- explain prime numbers
- explain LCM and GCF
- explain and memorize the rules for divisibility tests of numbers
- explain the concept of indices (powers) as shorthand for repeated multiplication of factors
- factor numbers using prime factors

Inquiry questions:

- How can prime factorization and divisibility tests help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Prime factorization and divisibility tests: Learn

Prime factorization and divisibility tests helps you break numbers into useful groups.

Core idea:

- A factor divides a number exactly.
- A multiple is made by multiplying a number by 1, 2, 3, and so on.
- Prime numbers have exactly two factors: 1 and the number itself.
- Classification is easier when you test divisibility carefully.

Key words:

- Prime factorization and divisibility tests
- Prime
- factorization
- divisibility
- tests
- method
- model
- unit

Prime factorization and divisibility tests: Worked Example

Worked example for Prime factorization and divisibility tests: List all factors of 24.

1. Test factor pairs: 1×24 , 2×12 , 3×8 , 4×6 .
2. No whole-number factor pair is missing between 4 and 6.
3. Write the factors in order.

Answer: 1, 2, 3, 4, 6, 8, 12, 24.

Prime factorization and divisibility tests: Vocabulary And Study Skill

Use these words and study moves before you practise Prime factorization and divisibility tests. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Prime factorization and divisibility tests
- Prime
- factorization
- divisibility
- tests
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Prime factorization and divisibility tests without a clear example, method, or evidence.

Prime factorization and divisibility tests: Guided Activity

Guided activity for Prime factorization and divisibility tests:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Prime factorization and divisibility tests: Practice And Correction

Practice for Prime factorization and divisibility tests:

Main outcome: prime factorize, exploit the rule of divisibility tests less than 10, find the Lowest Common Multiple (LCM) and the Greatest Common Factor (GCF) of whole numbers.

1. Solve one prime factorization and divisibility tests question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to prime factorization and divisibility tests and solve it.

Prime factorization and divisibility tests: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Prime factorization and divisibility tests.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Prime factorization and divisibility tests: Outcome Check

Outcome check for Prime factorization and divisibility tests: Prime factorize, exploit the rule of divisibility tests less than 10, find the Lowest Common Multiple (LCM) and the Greatest Common Factor (GCF) of whole numbers.

1. List all factors of 18.
2. List the first five multiples of 6.
3. Circle any prime numbers in 2, 3, 4, 5, 6.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Prime factorization and divisibility tests: Outcome Check

Outcome check for Prime factorization and divisibility tests: Explain prime numbers.

1. List all factors of 18.
2. List the first five multiples of 6.
3. Circle any prime numbers in 2, 3, 4, 5, 6.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Equivalent fractions and operations: Lesson Opener

Learning focus: Equivalent fractions and operations

Country link: a Rwandan school, market, farming, building, transport, or data problem for Equivalent fractions and operations.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- add, subtract and find equivalent fractions and apply this knowledge to solve related problems
- explain the concept of equivalent fractions
- explain how to find equivalent fractions
- give examples of equivalent fractions
- explain how to add or subtract fractions with different denominators using equivalent fractions and the Lowest Common Multiple (LCM)
- shade equivalent fractions

Inquiry questions:

- How can equivalent fractions and operations help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Equivalent fractions and operations: Learn

Equivalent fractions and operations teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Equivalent fractions and operations
- Equivalent
- fractions
- operations
- method
- model
- unit
- answer

Equivalent fractions and operations: Worked Example

Worked example for Equivalent fractions and operations: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Equivalent fractions and operations: Vocabulary And Study Skill

Use these words and study moves before you practise Equivalent fractions and operations. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Equivalent fractions and operations
- Equivalent
- fractions
- operations
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Equivalent fractions and operations without a clear example, method, or evidence.

Equivalent fractions and operations: Guided Activity

Guided activity for Equivalent fractions and operations:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Equivalent fractions and operations: Practice And Correction

Practice for Equivalent fractions and operations:

Main outcome: add, subtract and find equivalent fractions and apply this knowledge to solve related problems.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to equivalent fractions and operations and solve it.

Equivalent fractions and operations: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Equivalent fractions and operations.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Equivalent fractions and operations: Outcome Check

Outcome check for Equivalent fractions and operations: Add, subtract and find equivalent fractions and apply this knowledge to solve related problems.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Equivalent fractions and operations: Outcome Check

Outcome check for Equivalent fractions and operations: Explain the concept of equivalent fractions.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplication and division of decimal numbers: Lesson Opener

Learning focus: Multiplication and division of decimal numbers

Country link: a Rwandan school, market, farming, building, transport, or data problem for Multiplication and division of decimal numbers.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- multiply, divide and compare decimal numbers up to 3 decimal places and apply this knowledge to solve related problems
- explain the concept of decimal numbers using place values up to three (3) decimal places
- identify the place values in written decimals numbers
- explain how to multiply and divide decimal numbers
- match fractions to decimals
- compare 2 or more decimal numbers using $>$, $<$ or $=$

Inquiry questions:

- How can multiplication and division of decimal numbers help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplication and division of decimal numbers: Learn

Multiplication and division of decimal numbers teaches another way to write parts of a whole using place value.

Core idea:

- The first digit after the decimal point shows tenths.
- The second digit shows hundredths.
- Decimal numbers are compared from left to right, just like whole numbers.
- Money and measurements often use decimals.

Key words:

- Multiplication and division of decimal numbers
- Multiplication
- division
- decimal
- numbers
- method
- model
- unit

Multiplication and division of decimal numbers: Worked Example

Worked example for Multiplication and division of decimal numbers: Compare 4.35 and 4.5.

1. Write 4.5 as 4.50 so both numbers show hundredths.
2. Compare whole numbers: both are 4.
3. Compare tenths: 3 tenths is less than 5 tenths.

Answer: 4.35 is less than 4.5.

Multiplication and division of decimal numbers: Vocabulary And Study Skill

Use these words and study moves before you practise Multiplication and division of decimal numbers. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Multiplication and division of decimal numbers
- Multiplication
- division
- decimal
- numbers
- method
- model
- unit

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Multiplication and division of decimal numbers without a clear example, method, or evidence.

Multiplication and division of decimal numbers: Guided Activity

Guided activity for Multiplication and division of decimal numbers:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Multiplication and division of decimal numbers: Practice And Correction

Practice for Multiplication and division of decimal numbers:

Main outcome: multiply, divide and compare decimal numbers up to 3 decimal places and apply this knowledge to solve related problems.

1. Write 3 tenths as a decimal.
2. Order 2.05, 2.5, and 2.15 from smallest to largest.
3. Use a money or measurement example to compare two decimal numbers.

Answer check:

Answers: 3 tenths = 0.3. Order: 2.05, 2.15, 2.5.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to multiplication and division of decimal numbers and solve it.

Multiplication and division of decimal numbers: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Multiplication and division of decimal numbers.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplication and division of decimal numbers: Outcome Check

Outcome check for Multiplication and division of decimal numbers: Multiply, divide and compare decimal numbers up to 3 decimal places and apply this knowledge to solve related problems.

1. Write 6 tenths as a decimal.
2. Compare 3.08 and 3.8 using place value.
3. Explain why your comparison is correct.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Multiplication and division of decimal numbers: Outcome Check

Outcome check for Multiplication and division of decimal numbers: Explain the concept of decimal numbers using place values up to three (3) decimal places.

1. Write 6 tenths as a decimal.
2. Compare 3.08 and 3.8 using place value.
3. Explain why your comparison is correct.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Application of direct proportions: Lesson Opener

Learning focus: Application of direct proportions

Country link: a Rwandan school, market, farming, building, transport, or data problem for Application of direct proportions.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- apply direct proportions in a practical context
- explain the concept of direct proportions
- explain how to work out problems involving direct proportions
- state where to apply direct proportions in a practical context
- calculate involving direct proportions
- solve mathematical problems that involve direct proportions

Inquiry questions:

- How can application of direct proportions help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Application of direct proportions: Learn

Application of direct proportions teaches how two quantities change together at the same rate.

Core idea:

- In direct proportion, when one quantity is multiplied, the other quantity is multiplied by the same factor.
- A table helps you compare matching values.
- The unit rate tells the value for one item, one litre, one metre, one hour, or one learner.
- Always check that the relationship is consistent before using proportion.

Key words:

- Application of direct proportions
- Application
- direct
- proportions
- method
- model
- unit
- answer

Application of direct proportions: Worked Example

Worked example for Application of direct proportions: If 3 exercise books cost 900 francs, how much do 5 exercise books cost at the same price?

1. Find the cost of 1 book: $900 / 3 = 300$ francs.
2. Multiply by 5 books: $300 \times 5 = 1500$ francs.
3. Check: more books should cost more money.

Answer: 5 exercise books cost 1500 francs.

Application of direct proportions: Vocabulary And Study Skill

Use these words and study moves before you practise Application of direct proportions. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Application of direct proportions
- Application
- direct
- proportions
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Application of direct proportions without a clear example, method, or evidence.

Application of direct proportions: Guided Activity

Guided activity for Application of direct proportions:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Application of direct proportions: Practice And Correction

Practice for Application of direct proportions:

Main outcome: apply direct proportions in a practical context.

1. Complete the table: 1 pen = 200 francs, 2 pens = __, 5 pens = __.
2. A cyclist travels 18 km in 2 hours at the same speed. How far in 5 hours?
3. Write the unit rate used in each answer.

Answer check:

Answers: 2 pens = 400 francs; 5 pens = 1000 francs. The cyclist travels 45 km in 5 hours because $18 / 2 = 9$ km per hour.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to application of direct proportions and solve it.

Application of direct proportions: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Application of direct proportions.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Application of direct proportions: Outcome Check

Outcome check for Application of direct proportions: Apply direct proportions in a practical context.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Application of direct proportions: Outcome Check

Outcome check for Application of direct proportions: Explain the concept of direct proportions.

1. A recipe uses 2 cups of flour for 6 cakes. How many cups for 18 cakes?
2. Write the unit rate or scale factor.
3. Check that both quantities changed by the same factor.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving measurements of length, capacity and mass: Lesson Opener

Learning focus: Solving problems involving measurements of length, capacity and mass

Country link: a Rwandan school, market, farming, building, transport, or data problem for Solving problems involving measurements of length, capacity and mass.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve problems involving measurements of length, capacity, mass and calculating number of intervals
- explain how to convert units of length, capacity and mass measurements
- define the relationship between the number of poles and the intervals on a closed or open line
- explain steps to follow in order to solve problems involving length, capacity and mass
- convert units of length, capacity and mass measurement
- calculate the number of intervals

Inquiry questions:

- How can solving problems involving measurements of length, capacity and mass help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving measurements of length, capacity and mass: Learn

Solving problems involving measurements of length, capacity and mass teaches how heavy objects are.

Core idea:

- Mass is measured with units such as grams and kilograms.
- A balance or scale helps compare mass fairly.
- Convert units before adding or comparing if the units are different.

Key words:

- Solving problems involving measurements of length, capacity and mass
- Solving
- problems
- involving
- measurements
- length
- capacity
- mass

Solving problems involving measurements of length, capacity and mass: Worked Example

Worked example for Solving problems involving measurements of length, capacity and mass: A bag of beans has a mass of 3 kg. A small packet has a mass of 500 g. What is the total mass in grams?

1. Convert 3 kg to grams: $3 \text{ kg} = 3000 \text{ g}$.
2. Add $3000 \text{ g} + 500 \text{ g} = 3500 \text{ g}$.

Answer: 3500 g.

Solving problems involving measurements of length, capacity and mass: Vocabulary And Study Skill

Use these words and study moves before you practise Solving problems involving measurements of length, capacity and mass. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Solving problems involving measurements of length, capacity and mass
- Solving
- problems
- involving
- measurements
- length
- capacity
- mass

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Solving problems involving measurements of length, capacity and mass without a clear example, method, or evidence.

Solving problems involving measurements of length, capacity and mass: Guided Activity

Guided activity for Solving problems involving measurements of length, capacity and mass:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Solving problems involving measurements of length, capacity and mass: Practice And Correction

Practice for Solving problems involving measurements of length, capacity and mass:

Main outcome: solve problems involving measurements of length, capacity, mass and calculating number of intervals.

1. Name two objects measured in grams and two measured in kilograms.
2. Convert 2 kg to grams.
3. Compare 1500 g and 2 kg. Which is heavier?

Answer check:

Answer: 2 kg = 2000 g, so 2 kg is heavier than 1500 g.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to solving problems involving measurements of length, capacity and mass and solve it.

Solving problems involving measurements of length, capacity and mass: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Solving problems involving measurements of length, capacity and mass.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving measurements of length, capacity and mass: Outcome Check

Outcome check for Solving problems involving measurements of length, capacity and mass: Solve problems involving measurements of length, capacity, mass and calculating number of intervals.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving measurements of length, capacity and mass: Outcome Check

Outcome check for Solving problems involving measurements of length, capacity and mass: Explain how to convert units of length, capacity and mass measurements.

1. Convert 4 kg to grams.
2. Compare 2500 g and 3 kg.
3. Write the heavier mass with a reason.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving time intervals: Lesson Opener

Learning focus: Solving problems involving time intervals

Country link: a Rwandan school, market, farming, building, transport, or data problem for Solving problems involving time intervals.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- solve real life problems that involve finding time intervals and conversion of units
- explain the various units of time
- state the units used in time measurements
- describe how to solve mathematical problems involving time
- explain the meaning of time intervals
- convert the measurement of time

Inquiry questions:

- How can solving problems involving time intervals help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving time intervals: Learn

Solving problems involving time intervals teaches how to read, write, convert, and use time.

Core idea:

- Time can be shown on analogue clocks, digital clocks, calendars, and timetables.
- 60 seconds make 1 minute, and 60 minutes make 1 hour.
- Timetables help you find start time, end time, and duration.

Key words:

- Solving problems involving time intervals
- Solving
- problems
- involving
- time
- intervals
- method
- model

Solving problems involving time intervals: Worked Example

Worked example for Solving problems involving time intervals: A lesson starts at 8:10 and ends at 8:50. How long is the lesson?

1. Count from 8:10 to 8:50.
2. 10 minutes to 8:20, 20 more to 8:40, 10 more to 8:50.
3. Total time = 40 minutes.

Answer: 40 minutes.

Solving problems involving time intervals: Vocabulary And Study Skill

Use these words and study moves before you practise Solving problems involving time intervals. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Solving problems involving time intervals
- Solving
- problems
- involving
- time
- intervals
- method
- model

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Solving problems involving time intervals without a clear example, method, or evidence.

Solving problems involving time intervals: Guided Activity

Guided activity for Solving problems involving time intervals:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Solving problems involving time intervals: Practice And Correction

Practice for Solving problems involving time intervals:

Main outcome: solve real life problems that involve finding time intervals and conversion of units.

1. Read a clock showing 7:30 and write the time in words.
2. Convert 2 hours to minutes.
3. A trip starts at 9:15 and ends at 10:00. Find the duration.

Answer check:

Answers: 2 hours = 120 minutes. 9:15 to 10:00 is 45 minutes.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to solving problems involving time intervals and solve it.

Solving problems involving time intervals: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Solving problems involving time intervals.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving time intervals: Outcome Check

Outcome check for Solving problems involving time intervals: Solve real life problems that involve finding time intervals and conversion of units.

1. Convert 3 hours to minutes.
2. Find the duration from 8:20 to 9:05.
3. Write the answer in minutes.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Solving problems involving time intervals: Outcome Check

Outcome check for Solving problems involving time intervals: Explain the various units of time.

1. Convert 3 hours to minutes.
2. Find the duration from 8:20 to 9:05.
3. Write the answer in minutes.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Money and its financial applications: Lesson Opener

Learning focus: Money and its financial applications

Country link: a Rwandan school, market, farming, building, transport, or data problem for Money and its financial applications.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- explain money and its financial applications
- explain how to changes money and give examples
- define the following terms: Buying price, selling price, profit, and loss
- receive an amount of money and provide the changes of using coins and notes
- calculate the selling price, buying price
- compare the buying price and the selling price and decide if there is a loss or a profit

Inquiry questions:

- How can money and its financial applications help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Money and its financial applications: Learn

Money and its financial applications teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Money and its financial applications
- Money
- financial
- applications
- method
- model
- unit
- answer

Money and its financial applications: Worked Example

Worked example for Money and its financial applications: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Money and its financial applications: Vocabulary And Study Skill

Use these words and study moves before you practise Money and its financial applications. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Money and its financial applications
- Money
- financial
- applications
- method
- model
- unit
- answer

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Money and its financial applications without a clear example, method, or evidence.

Money and its financial applications: Guided Activity

Guided activity for Money and its financial applications:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Money and its financial applications: Practice And Correction

Practice for Money and its financial applications:

Main outcome: explain money and its financial applications.

1. Solve one money and its financial applications question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to money and its financial applications and solve it.

Money and its financial applications: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Money and its financial applications.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Money and its financial applications: Outcome Check

Outcome check for Money and its financial applications: Explain money and its financial applications.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Money and its financial applications: Outcome Check

Outcome check for Money and its financial applications: Explain how to changes money and give examples.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Sequences that include whole numbers, fractions and decimals: Lesson Opener

Learning focus: Sequences that include whole numbers, fractions and decimals

Country link: a Rwandan school, market, farming, building, transport, or data problem for Sequences that include whole numbers, fractions and decimals.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- write sequences of whole numbers, fractions and decimals
- explain how to arrange whole numbers in increasing and decreasing order
- explain sequences of whole numbers, fractions, and decimals
- show the patterns used
- find the missing numbers in the sequence
- complete the consecutive terms of a sequence with a constant difference

Inquiry questions:

- How can sequences that include whole numbers, fractions and decimals help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Sequences that include whole numbers, fractions and decimals: Learn

Sequences that include whole numbers, fractions and decimals teaches another way to write parts of a whole using place value.

Core idea:

- The first digit after the decimal point shows tenths.
- The second digit shows hundredths.
- Decimal numbers are compared from left to right, just like whole numbers.
- Money and measurements often use decimals.

Key words:

- Sequences that include whole numbers, fractions and decimals
- Sequences
- that
- include
- whole
- numbers
- fractions
- decimals

Sequences that include whole numbers, fractions and decimals: Worked Example

Worked example for Sequences that include whole numbers, fractions and decimals: Compare 4.35 and 4.5.

1. Write 4.5 as 4.50 so both numbers show hundredths.
2. Compare whole numbers: both are 4.
3. Compare tenths: 3 tenths is less than 5 tenths.

Answer: 4.35 is less than 4.5.

Sequences that include whole numbers, fractions and decimals: Vocabulary And Study Skill

Use these words and study moves before you practise Sequences that include whole numbers, fractions and decimals. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Sequences that include whole numbers, fractions and decimals
- Sequences
- that
- include
- whole
- numbers
- fractions
- decimals

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Sequences that include whole numbers, fractions and decimals without a clear example, method, or evidence.

Sequences that include whole numbers, fractions and decimals: Guided Activity

Guided activity for Sequences that include whole numbers, fractions and decimals:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Sequences that include whole numbers, fractions and decimals: Practice And Correction

Practice for Sequences that include whole numbers, fractions and decimals:

Main outcome: write sequences of whole numbers, fractions and decimals.

1. Write 3 tenths as a decimal.
2. Order 2.05, 2.5, and 2.15 from smallest to largest.
3. Use a money or measurement example to compare two decimal numbers.

Answer check:

Answers: 3 tenths = 0.3. Order: 2.05, 2.15, 2.5.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to sequences that include whole numbers, fractions and decimals and solve it.

Sequences that include whole numbers, fractions and decimals: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Sequences that include whole numbers, fractions and decimals.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Sequences that include whole numbers, fractions and decimals: Outcome Check

Outcome check for Sequences that include whole numbers, fractions and decimals: Write sequences of whole numbers, fractions and decimals.

1. Write 6 tenths as a decimal.
2. Compare 3.08 and 3.8 using place value.
3. Explain why your comparison is correct.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Sequences that include whole numbers, fractions and decimals: Outcome Check

Outcome check for Sequences that include whole numbers, fractions and decimals: Explain how to arrange whole numbers in increasing and decreasing order.

1. Write 6 tenths as a decimal.
2. Compare 3.08 and 3.8 using place value.
3. Explain why your comparison is correct.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Constructing angles and describing angles formed by intersecting lines: Lesson Opener

Learning focus: Constructing angles and describing angles formed by intersecting lines

Country link: a Rwandan school, market, farming, building, transport, or data problem for Constructing angles and describing angles formed by intersecting lines.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- construct different angles and describe properties of angles formed by intersecting lines
- explain the concept of parallel, transversal, intersecting, and perpendicular lines
- describe the process of drawing angles and the process of constructing angles using a ruler and a pair of compasses
- measure angles using a protractor
- construct angles using a ruler and a pair of compasses
- carry out and explain the process of bisection of an angle using a ruler and compass

Inquiry questions:

- How can constructing angles and describing angles formed by intersecting lines help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Constructing angles and describing angles formed by intersecting lines: Learn

Constructing angles and describing angles formed by intersecting lines teaches how lines and angles describe shapes and directions.

Core idea:

- A line can be straight, parallel, perpendicular, horizontal, vertical, or slanting.
- An angle is a turn between two lines.
- Draw and label diagrams carefully before measuring or classifying.

Key words:

- Constructing angles and describing angles formed by intersecting lines
- Constructing
- angles
- describing
- formed
- intersecting
- lines
- method

Constructing angles and describing angles formed by intersecting lines: Worked Example

Worked example for Constructing angles and describing angles formed by intersecting lines: A shape has one right angle. How can you show it clearly?

1. Draw the two meeting lines.
2. Mark the square corner to show a right angle.
3. Label it 90 degrees.
4. Name a classroom example, such as a book corner.

Answer: A right angle is a quarter turn, 90 degrees.

Constructing angles and describing angles formed by intersecting lines: Vocabulary And Study Skill

Use these words and study moves before you practise Constructing angles and describing angles formed by intersecting lines. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Constructing angles and describing angles formed by intersecting lines
- Constructing
- angles
- describing
- formed
- intersecting
- lines
- method

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Constructing angles and describing angles formed by intersecting lines without a clear example, method, or evidence.

Constructing angles and describing angles formed by intersecting lines: Guided Activity

Guided activity for Constructing angles and describing angles formed by intersecting lines:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Constructing angles and describing angles formed by intersecting lines: Practice And Correction

Practice for Constructing angles and describing angles formed by intersecting lines:

Main outcome: construct different angles and describe properties of angles formed by intersecting lines.

1. Draw one right angle and label it 90 degrees.
2. Name one pair of parallel lines in the classroom.
3. Use a ruler to draw a straight line segment and label its endpoints.

Answer check:

Answer check: the right angle should show a square corner; parallel lines stay the same distance apart.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to constructing angles and describing angles formed by intersecting lines and solve it.

Constructing angles and describing angles formed by intersecting lines: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Constructing angles and describing angles formed by intersecting lines.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Constructing angles and describing angles formed by intersecting lines: Outcome Check

Outcome check for Constructing angles and describing angles formed by intersecting lines: Construct different angles and describe properties of angles formed by intersecting lines.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Constructing angles and describing angles formed by intersecting lines: Outcome Check

Outcome check for Constructing angles and describing angles formed by intersecting lines: Explain the concept of parallel, transversal, intersecting, and perpendicular lines.

1. Draw and label one right angle.
2. Name one pair of parallel lines.
3. Explain how you know.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Interpreting and constructing scale drawings: Lesson Opener

Learning focus: Interpreting and constructing scale drawings

Country link: a Rwandan school, market, farming, building, transport, or data problem for Interpreting and constructing scale drawings.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- interpret and construct scale drawings
- explain the concept of scale drawing
- explain how to find the actual distance on the ground
- explain how to find the scale of drawings and maps
- calculate the scale of a map
- calculate the actual distance on the ground

Inquiry questions:

- How can interpreting and constructing scale drawings help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Interpreting and constructing scale drawings: Learn

Interpreting and constructing scale drawings teaches a mathematical idea by using a clear model, a labelled representation, step-by-step working, and a final check.

Key words:

- Interpreting and constructing scale drawings
- Interpreting
- constructing
- scale
- drawings
- method
- model
- unit

Interpreting and constructing scale drawings: Worked Example

Worked example for Interpreting and constructing scale drawings: Read the task, choose a representation, solve one step at a time, and check the answer in context.

Model:

1. Name what is known.
2. Choose a diagram, table, equation, or model.
3. Complete the calculation or reasoning.
4. Check the result with estimation or inverse reasoning.

Use a Rwanda classroom, market, garden, transport, or home example.

Interpreting and constructing scale drawings: Vocabulary And Study Skill

Use these words and study moves before you practise Interpreting and constructing scale drawings. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Interpreting and constructing scale drawings
- Interpreting
- constructing
- scale
- drawings
- method

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Interpreting and constructing scale drawings without a clear example, method, or evidence.

Interpreting and constructing scale drawings: Guided Activity

Guided activity for Interpreting and constructing scale drawings:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Interpreting and constructing scale drawings: Practice And Correction

Practice for Interpreting and constructing scale drawings:

Main outcome: interpret and construct scale drawings.

1. Solve one interpreting and constructing scale drawings question using labelled steps.
2. Show your method with a drawing, table, number sentence, or diagram.
3. Check your answer and write one sentence explaining it.

Answer check:

Your answer should show a clear model, correct method, correct labels, and a final sentence.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to interpreting and constructing scale drawings and solve it.

Interpreting and constructing scale drawings: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Interpreting and constructing scale drawings.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Interpreting and constructing scale drawings: Outcome Check

Outcome check for Interpreting and constructing scale drawings: Interpret and construct scale drawings.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Interpreting and constructing scale drawings: Outcome Check

Outcome check for Interpreting and constructing scale drawings: Explain the concept of scale drawing.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Calculating circumference of a circle, surface area and volume of cubes and cuboids: Lesson Opener

Learning focus: Calculating circumference of a circle, surface area and volume of cubes and cuboids

Country link: a Rwandan school, market, farming, building, transport, or data problem for Calculating circumference of a circle, surface area and volume of cubes and cuboids.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- calculate circumference of a circle, surface area and volume of cubes and cuboids
- explain the concept of circumference
- describe the process of finding the circumference of a circle
- establish the relationships between cubes and cuboids
- explain how to find the volume of cubes or cuboids
- practically calculate the value of pi

Inquiry questions:

- How can calculating circumference of a circle, surface area and volume of cubes and cuboids help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Calculating circumference of a circle, surface area and volume of cubes and cuboids: Learn

Calculating circumference of a circle, surface area and volume of cubes and cuboids teaches properties of three-dimensional shapes such as cubes, cuboids, and cylinders.

Core idea:

- Solid shapes have faces, edges, and vertices.
- A cuboid has rectangular faces.
- A cube has six equal square faces.
- A cylinder has two circular faces and one curved surface.
- Nets show how flat faces fold to make a solid.

Key words:

- Calculating circumference of a circle, surface area and volume of cubes and cuboids
- Calculating
- circumference
- circle
- surface
- area
- volume
- cubes

Calculating circumference of a circle, surface area and volume of cubes and cuboids: Worked Example

Worked example for Calculating circumference of a circle, surface area and volume of cubes and cuboids: Name the faces of a cuboid.

1. Look at the solid shape.
2. A cuboid has 6 faces.
3. Each face is a rectangle.
4. Opposite faces are equal in size.

Answer: A cuboid has 6 rectangular faces, 12 edges, and 8 vertices.

Calculating circumference of a circle, surface area and volume of cubes and cuboids: Vocabulary And Study Skill

Use these words and study moves before you practise Calculating circumference of a circle, surface area and volume of cubes and cuboids. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Calculating circumference of a circle, surface area and volume of cubes and cuboids
- Calculating
- circumference
- circle
- surface
- area
- volume
- cubes

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Calculating circumference of a circle, surface area and volume of cubes and cuboids without a clear example, method, or evidence.

Calculating circumference of a circle, surface area and volume of cubes and cuboids: Guided Activity

Guided activity for Calculating circumference of a circle, surface area and volume of cubes and cuboids:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Calculating circumference of a circle, surface area and volume of cubes and cuboids: Practice And Correction

Practice for Calculating circumference of a circle, surface area and volume of cubes and cuboids:

Main outcome: calculate circumference of a circle, surface area and volume of cubes and cuboids.

1. Draw or model a cube, cuboid, and cylinder.
2. Count faces, edges, and vertices where they apply.
3. Sketch a simple net for a cube or cuboid.

Answer check:

Answer check: cube has 6 equal square faces; cuboid has 6 rectangular faces; cylinder has circular faces and a curved surface.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to calculating circumference of a circle, surface area and volume of cubes and cuboids and solve it.

Calculating circumference of a circle, surface area and volume of cubes and cuboids: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Calculating circumference of a circle, surface area and volume of cubes and cuboids.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Calculating circumference of a circle, surface area and volume of cubes and cuboids: Outcome Check

Outcome check for Calculating circumference of a circle, surface area and volume of cubes and cuboids: Calculate circumference of a circle, surface area and volume of cubes and cuboids.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Calculating circumference of a circle, surface area and volume of cubes and cuboids: Outcome Check

Outcome check for Calculating circumference of a circle, surface area and volume of cubes and cuboids: Explain the concept of circumference.

1. Name the faces of a cuboid.
2. Count edges and vertices of a cube.
3. Describe one difference between a cuboid and a cylinder.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Data handling: Lesson Opener

Learning focus: Data handling

Country link: a Rwandan school, market, farming, building, transport, or data problem for Data handling.

Progression focus: add reasons, compare examples, and explain choices more independently; show methods, units, checks, and a final explanation.

By the end of this topic, you should be able to:

- collect data, represent and interpret it in order to answer a question or explore a hypothesis
- define continuous and discrete data
- explain how to construct bar charts
- explain line graphs
- distinguish continuous from discrete data
- collect data and record it in a table

Inquiry questions:

- How can data handling help you solve a real problem in Mathematics?

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Data handling: Learn

Data handling teaches how to collect, organize, read, and explain information.

Core idea:

- Data must answer a clear question.
- Tables, tally charts, pictographs, and bar charts make data easier to read.
- A good conclusion says what the data shows.

Key words:

- Data handling
- Data
- handling
- method
- model
- unit
- answer
- check

Data handling: Worked Example

Worked example for Data handling: A class records favourite fruits: mango 8, banana 12, pineapple 6. Which fruit is most popular?

1. Read the table values.
2. Compare 8, 12, and 6.
3. 12 is the greatest number.

Answer: Banana is most popular.

Data handling: Vocabulary And Study Skill

Use these words and study moves before you practise Data handling. Good vocabulary helps you read the question, choose the right method, and explain your answer clearly.

Key words:

- Data handling
- Data
- handling
- method
- model
- unit
- answer
- check

Study skill:

1. Choose three key words.
2. Say each word in your own words.
3. Give one example from Rwanda.
4. Use each word in a correct sentence, calculation step, diagram label, or oral answer.
5. Ask a partner to check whether your example matches the topic.

Common mistake to avoid: giving a general answer about Data handling without a clear example, method, or evidence.

Data handling: Guided Activity

Guided activity for Data handling:

1. Work with a partner and read the worked example.
2. Make a similar question using Rwanda classroom, home, market, garden, transport, or timetable details.
3. Draw a model: table, number line, diagram, place-value chart, shape, or equation.
4. Solve the question step by step.
5. Swap work with another pair and check the method, units, labels, and final sentence.

Teacher check: the model should match the topic, not just any calculation.

Data handling: Practice And Correction

Practice for Data handling:

Main outcome: collect data, represent and interpret it in order to answer a question or explore a hypothesis.

1. Ask 10 learners a simple question and make a tally table.
2. Draw a bar chart from the tally table.
3. Write one conclusion from the data.

Answer check:

Answer check: totals in the tally table and bar chart should match.

Correction check:

- Did you show the method?
- Did you use the correct unit or label?
- Did you check whether the answer is reasonable?

Home link: Find one Rwanda home, school, shop, garden, road, game, or timetable example connected to data handling and solve it.

Data handling: Challenge And Remediation

Use this page after practice.

If you need support:

1. Re-read the success criteria.
2. Copy the worked example steps as headings.
3. Replace the model details with details from Data handling.
4. Ask one clear question about the step that is confusing.

If you are ready for a challenge:

1. Create a new task from school, home, community, environment, technology, sport, art, market, or fieldwork.
2. Solve or answer it using correct vocabulary.
3. Explain why your answer is reasonable.
4. Write one extension question for a classmate.

Check:

- method is clear
- working is shown step by step
- answer is checked in context

Data handling: Outcome Check

Outcome check for Data handling: Collect data, represent and interpret it in order to answer a question or explore a hypothesis.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Data handling: Outcome Check

Outcome check for Data handling: Define continuous and discrete data.

1. Write one clear example from the topic.
2. Solve it with labelled steps.
3. Check the answer in context.

Success criteria:

- method is clear
- working is shown step by step
- answer is checked in context

Reading, writing, comparing and calculating whole numbers up to 1,000,000: Review Clinic 1

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Reading, writing, comparing and calculating whole numbers up to 1,000,000 that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Addition and subtraction of integers: Review Clinic 2

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Addition and subtraction of integers that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Prime factorization and divisibility tests: Review Clinic 3

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Prime factorization and divisibility tests that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Equivalent fractions and operations: Review Clinic 4

Use this review clinic to strengthen Mathematics without copying earlier answers.

1. Choose one idea from Equivalent fractions and operations that was difficult.
2. Explain the idea in your own words.
3. Make one new example from Rwanda.
4. Create one question that checks knowledge.
5. Create one question that checks skill or application.
6. Answer both questions and correct your work.

Challenge: Turn the idea into a drawing, table, map, dialogue, model, demonstration, or worked solution.

Glossary

Use this glossary to revise important words in Mathematics.

- Reading, writing, comparing and calculating whole numbers up to 1,000,000: Explain this term using an example from the book, your school, or your community.
- Addition and subtraction of integers: Explain this term using an example from the book, your school, or your community.
- Prime factorization and divisibility tests: Explain this term using an example from the book, your school, or your community.
- Equivalent fractions and operations: Explain this term using an example from the book, your school, or your community.
- Multiplication and division of decimal numbers: Explain this term using an example from the book, your school, or your community.
- Application of direct proportions: Explain this term using an example from the book, your school, or your community.
- Solving problems involving measurements of length, capacity and mass: Explain this term using an example from the book, your school, or your community.
- Solving problems involving time intervals: Explain this term using an example from the book, your school, or your community.
- Money and its financial applications: Explain this term using an example from the book, your school, or your community.
- Sequences that include whole numbers, fractions and decimals: Explain this term using an example from the book, your school, or your community.
- Constructing angles and describing angles formed by intersecting lines: Explain this term using an example from the book, your school, or your community.
- Interpreting and constructing scale drawings: Explain this term using an example from the book, your school, or your community.
- Calculating circumference of a circle, surface area and volume of cubes and cuboids: Explain this term using an example from the book, your school, or your community.
- Data handling: Explain this term using an example from the book, your school, or your community.

Add five more words from your lessons and write your own meanings.

Answer And Teacher Notes

A strong Mathematics answer shows the model, method, units or labels, final answer, and a check. Use the worked examples and answer checks in each practice page to correct your work. When an answer is wrong, find the first step that changed the meaning and correct that step before trying a new question.

Final Project

Create a final project for Mathematics that shows what you can now do independently.

Your project must include:

1. A clear title.
2. A real-life problem, question, text, performance, investigation, design, or worked task.
3. Evidence from at least three lessons in this book.
4. A drawing, table, map, chart, model, dialogue, paragraph, performance plan, practical product, or worked solution.
5. A correction note showing one improvement after feedback.
6. A short reflection explaining what you learned, what was difficult, and what you will practise next.

Presentation check: keep the work neat, label important parts, use correct vocabulary, and be ready to explain your choices to a classmate or teacher.